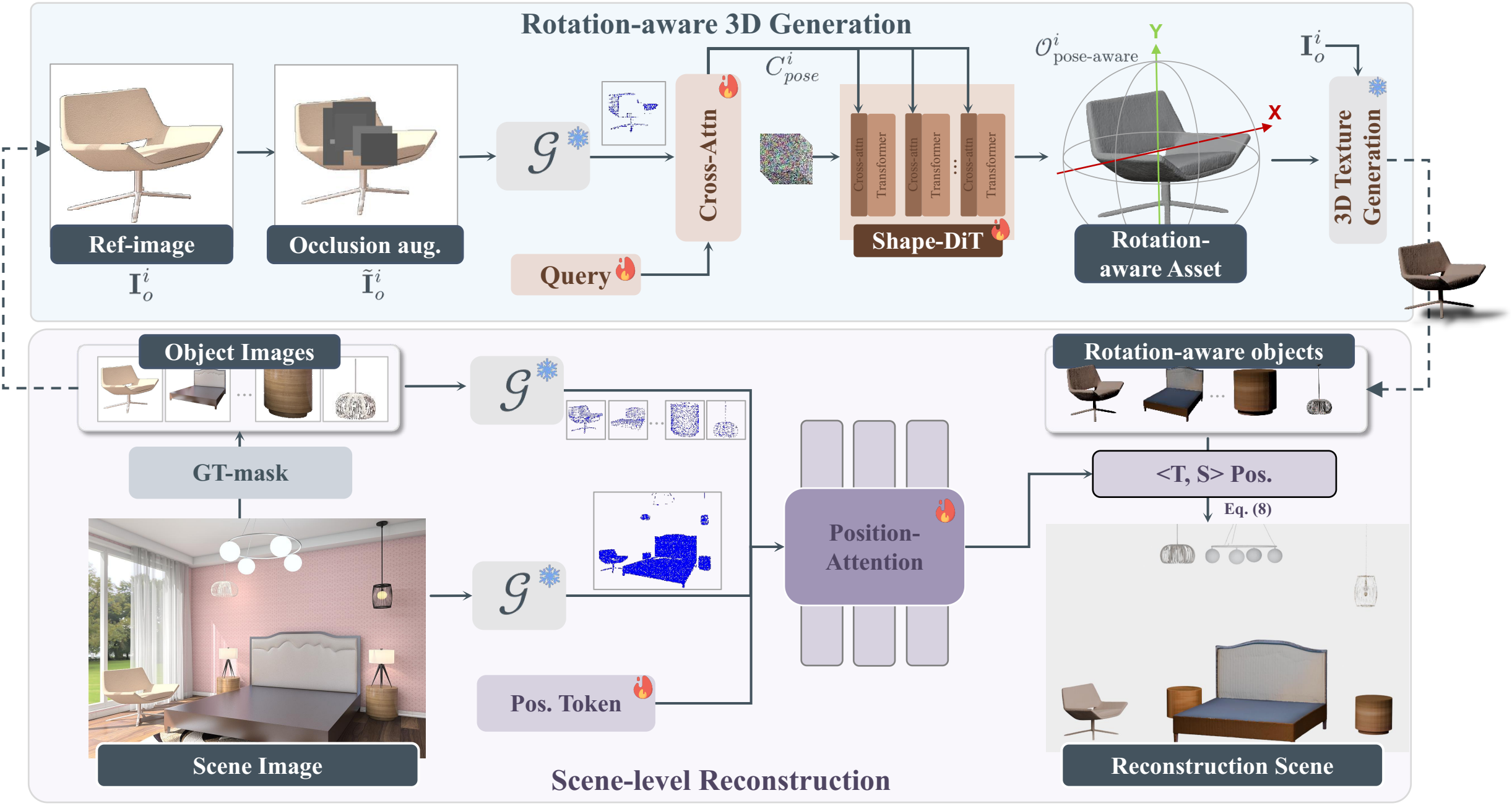

Rotation-aware 3D Generation
Ref-image
$\mathbf{I}_o^i$
Occlusion aug.
$\tilde{\mathbf{I}}_o^i$
$\mathcal{G}$
Query
Cross-Attn
C_{pose}^i
Cross-attn
Transformer
Shape-DiT
$\mathcal{O}_{\text{pose-aware}}^i$
Y
X
Rotation-aware Asset
$\mathbf{I}_o^i$
3D Texture Generation
Object Images
GT-mask
Scene Image
Pos. Token
Position-Attention
Rotation-aware objects
<T, S> Pos.
Eq. (8)
Reconstruction Scene
Scene-level Reconstruction